MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 01
Chapter 10: Circles

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. From a point P which is at a distance of 13 cm from the centre O of a circle of radius 5 cm, the length of the tangent drawn to the circle is: [1]

(A) 10 cm (B) 11 cm (C) 12 cm (D) $\sqrt{194}$ cm

Q2. If two tangents TP and TQ are drawn to a circle with centre O such that $\angle POQ = 110°$, then the measure of $\angle PTQ$ is equal to: [1]

(A) 60° (B) 70° (C) 80° (D) 90°

Q3. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°, then $\angle POA$ is: [1]

(A) 50° (B) 60° (C) 70° (D) 80°

Q4. Two concentric circles are of radii 5 cm and 3 cm. The length of the chord of the larger circle which touches the smaller circle is: [1]

(A) 6 cm (B) 7 cm (C) 8 cm (D) 10 cm

Q5. **Assertion (A):** The lengths of two tangents drawn from an external point to a circle are equal.
Reason (R): The tangents drawn from an external point to a circle subtend equal angles at the centre. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Two concentric circles are of radii 6.5 cm and 2.5 cm. Find the length of the chord of the larger circle which touches the smaller circle. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Prove that the lengths of tangents drawn from an external point to a circle are equal. [3]

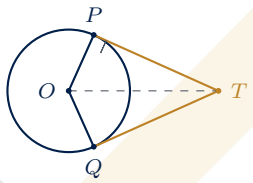
Q8. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$. [3]

Q9. A circle touches the side BC of $\triangle ABC$ at point P and touches AB and AC produced at Q and R respectively. Prove that $AQ = \frac{1}{2}(\text{Perimeter of } \triangle ABC)$. [3]

SECTION D: Case-Based / Applied Competency Question (1 × 4 = 4 Marks)

Q10. Industrial Pulley and Belt Drive Transmission

An engineer designs a mechanical conveyor system where a drive belt runs tautly around a circular drive wheel of radius $r = 5$ cm centred at O. The belt leaves the wheel tangentially at points P and Q and converges at an external tensioner bracket T located 13 cm from the wheel centre ($OT = 13$ cm).



(i) State the mathematical relationship between radius OP and tangent segment TP at the point of contact. [1]

(ii) Calculate the exact length of the tangent belt segment TP. [1]

(iii) Find the measure of central angle $\angle POQ$ if the angle between the belt lines $\angle PTQ = 45°$, OR calculate the area of quadrilateral $OPTQ$. [2]

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